Lesson 3.2 **Duration: 40 minutes**

Database Concepts

Learning Objectives

- Explain the concepts of tables, records, fields, and primary keys
- Describe how relationships connect data across tables
- Distinguish between data types used in database fields

Engage (5-7 minutes)

Think of a phone contact list: each contact is a row, and their name, number, and email are columns. Now imagine linking contacts to call history—each call record references a contact. This is how databases organize and connect information.

Explore (10-12 minutes)

Using a class register as a reference, identify the table name, list three records (rows), and list five fields (columns). Then determine which field uniquely identifies each student and explain why that field must be unique.

Explain (10-12 minutes)

A table is a grid of data with rows (records) and columns (fields). Each field has a data type—TEXT for names, INTEGER for roll numbers, DATE for birthdays, and BOOLEAN for yes/no values. A primary key uniquely identifies every record. Foreign keys create relationships by referencing the primary key of another table.

Elaborate (8-10 minutes)

Design a mini-database for a bookstore with two tables: Books and Authors. Write the fields for each table, identify the primary keys, and explain how you would link an author to their books using a foreign key relationship.

Evaluate (5-8 minutes)

Quiz: Given a table “Students” with columns ID, Name, Grade, and Phone—identify the primary key, suggest a suitable data type for each column, and explain what happens if two students share the same ID.